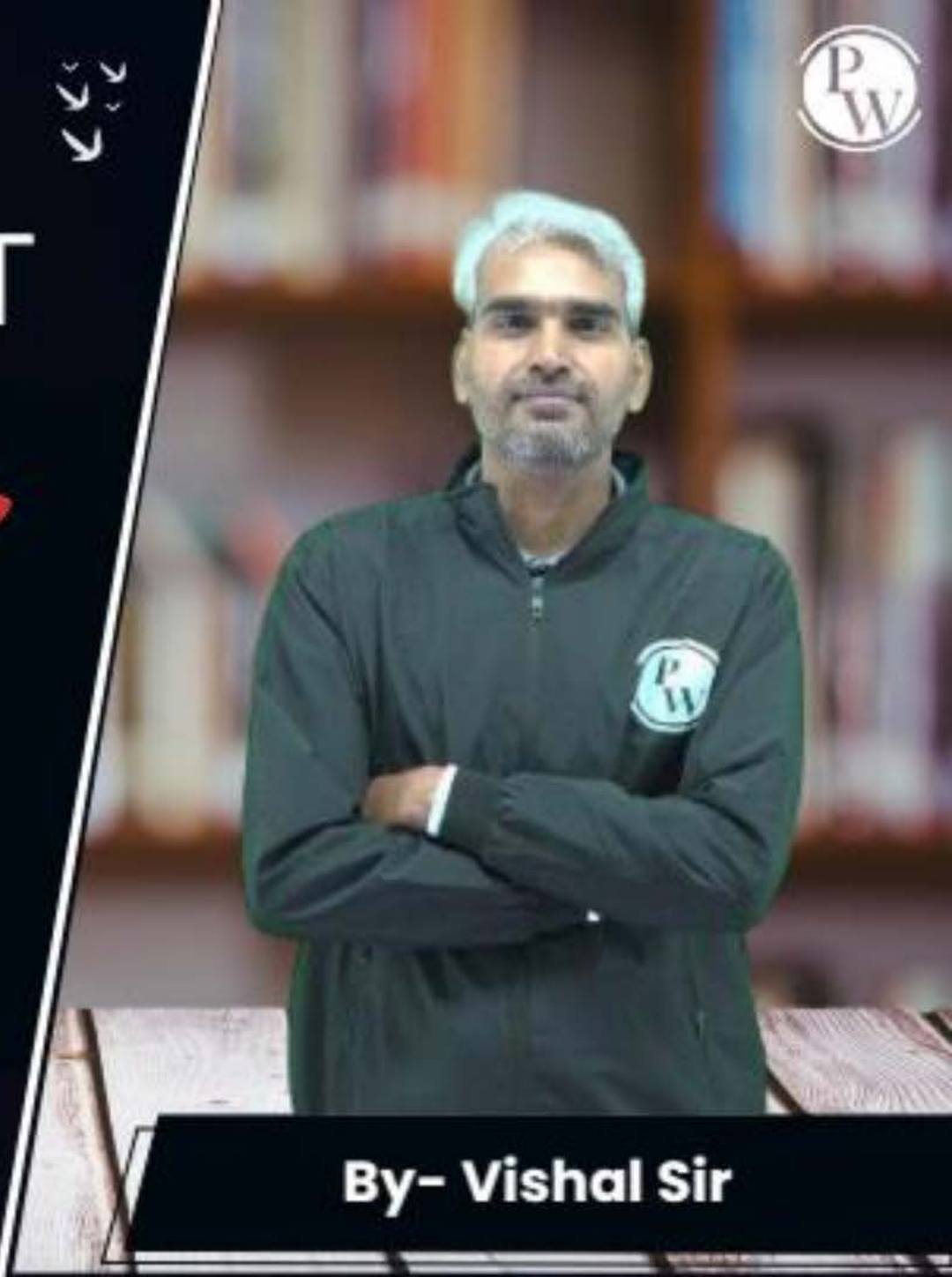
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 09



By- Vishal Sir

Recap of Previous Lecture



- Topic: Transitive relation
- Topic: Reflexive closure
- Topic: Symmetric closure
- Topic: Transitive closure

Slide

Topics to be Covered



- Topic — Transitive closure
- Topic — Complement of a relation
- Topic — Inverse of a relation
- Topic — Composite of two relations

Slide



Topic : Transitive Closure

- Let R is any relation on set A, Transitive closure of relation R is the smallest transitive relation on set A that contains relation R

Slide

Note:

If relation R on set A is already a transitive relation, then transitive Closure of relation R will be relation R itself.

eg 1: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

Find transitive closure of R

eg 1: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

Given, $R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

1st iteration → $R^* = \{(1,1), (1,2), (1,3), (2,2), (3,1) \mid (3,2) (3,3)\}$ ✓

2nd iteration → $R^{**} = \{(1,1), (1,2), (1,3), (2,2), (3,1), (3,2), (3,3)\}$

In this iteration, no new order pair needs to be added to satisfy transitivity

We will keep iterating, until we reach to a situation when no new order pair needs to be added in order to satisfy transitivity

And the relation w.r.t. last iteration (i.e. the iteration in which no new order pair is added) will be the transitive Closure of given relation

eg 2: Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$

Find transitive closure of R

$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$

1st iteration

$R^* = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$ ...

No new order pair needs to be added

Transitive closure of R is R^*

if $\boxed{R^* = R}$ i.e, given relation is a transitive relation

eg. Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$

Find transitive closure of R.

$R = \{(1,1)\ (1,3), (2,2), (3,1), (3,2)\}$

1st iteration → $R^* = \{(1,1)\ (1,3), (2,2), (3,1), (3,2) \mid (1,2)\ (3,3)\}$

Rearranged R^* (it is not iteration) $= \{(1,1)\ (1,2)\ (1,3), (2,2), (3,1), (3,2)\ (3,3)\}$

$R^{**} = \{(1,1)\ (1,2)\ (1,3), (2,2), (3,1), (3,2)\ (3,3) \mid$

No new order pair needs to be added

∴ R^{**} is transitive Closure of R

Today's Topic

H.W.

eg. Let $A = \{a, b, c, d\}$

$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

Find transitive closure of R

$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

$R^* = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c) \mid (a,c), (b,d), (c,c), (d,a), (d,d)\}$

$R^{**} = \{(a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d) \mid (a,a)\}$

$R^{***} = \{(a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)\}$

No new order pair needs to be added

R^{***} is the transitive closure of R.

{ Warshall's algo }

Warshall's algo can be used to find the transitive closure of given relation.

eg: Let $A = \{a, b, c, d\}$

$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

Find transitive closure of R

Matrix w.r.t relation R = ((x, y) → a, b, c, d)

$$\begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

	I st	II nd	III rd	IV th
Column	{b, c}	{ }	{b, d}	{a, c}
Row	{d}	{a, c}	{a, d}	{c}

Column x Row = $\{(b,d), (c,d)\}$ $\{\}$ $\{(b,a), (b,d), (d,a), (d,d)\}$ $\{(a,c), (c,c)\}$

$R^* = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)$
$(b,d), (d,a), (d,d), (a,c), (c,c)\}$

$R^* = \{(a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)\}$

Matrix w.r.t. relation R^* (entry (x,y) is 1 if $(x,y) \in R^*$):

	a	b	c	d
a	0	0	1	1
b	1	0	1	1
c	1	0	1	1
d	1	0	1	1

	I st	II nd	III rd	IV th
Column	{b, c, d}	{ }	{a, b, c, d}	{a, b, c, d}
Row	{c, d}	{a, c, d}	{a, c, d}	{a, c, d}
Column x Row =	(b,c) (b,d) (c,c) (c,d) (d,c) (d,d)		(a,a), (a,c) (a,d), (b,a) (b,c), (b,d) (c,a) (c,c) (c,d), (d,a) (d,c), (d,d)	(a,a), (a,c) (a,d), (b,a) (b,c), (b,d) (c,a) (c,c) (c,d), (d,a) (d,c), (d,d)

$R^{**} = \{(a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d), (a,a)\}$

$R^{**} = \{ (a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d) \}$

Matrix w.r.t. relation R^{**}:

(x, y) → a b c d

	a	b	c	d
a	1	0	1	1
b	1	0	1	1
c	1	0	1	1
d	1	0	1	1

	I st	II nd	III rd	IV th
Column	{a, b, c, d}	{ }	{a, b, c, d}	{a, b, c, d}
Row	{a, c, d}	{a, c, d}	{a, c, d}	{a, c, d}
Column x Row =	(a,a), (a,c) (a,d), (b,a) (b,c), (b,d) (c,a) (c,c) (c,d), (d,a) (d,c), (d,d)		(a,a), (a,c) (a,d), (b,a) (b,c), (b,d) (c,a) (c,c) (c,d), (d,a) (d,c), (d,d)	(a,a), (a,c) (a,d), (b,a) (b,c), (b,d) (c,a) (c,c) (c,d), (d,a) (d,c), (d,d)

$R^{***} = (a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)$

No new order pair is added

∴ R^{***} is transitive closure of R.



Topic : Complement of a relation

Complement is always defined w.r.t. Universal set

- Let 'R' is a relation from set A to set B, then the complement of relation R is denoted by R^c and it is defined as.

$$R^c = A \times B - R$$

$$A^c = U - A$$

$A \times B = \{(1,a), (1,b), (1,c), (2,a), (2,b), (2,c)\}$

eg. $A = \{1, 2\}$ $B = \{a, b, c\}$

and R is a relation from set A to set B,

$R = \{(1,a), (2,b), (2,c)\}$

then $R^c = \{(1,b), (1,c), (2,a)\}$

Slide



Topic : Inverse of a relation

- Let R is a relation from set A to set B.
  Inverse relation of relation R can be denoted by R^{-1}.
  and R^{-1} will be a relation from set B to set A
  which is defined by the rule

$$R^{-1} = \{(b,a) \mid (a,b) \in R\}$$

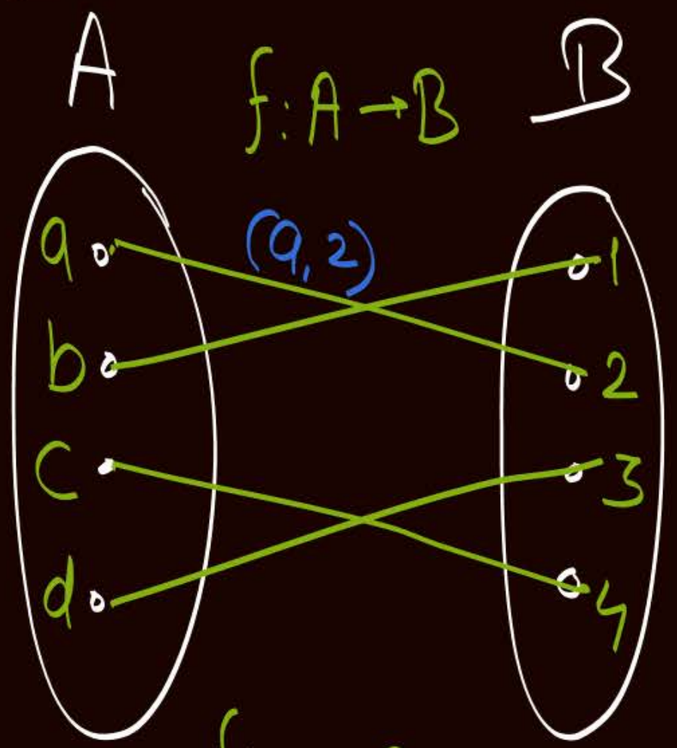
(b) from set B, (a) from set A, (a,b) ∈ R, (a) from set A, (b) from set B

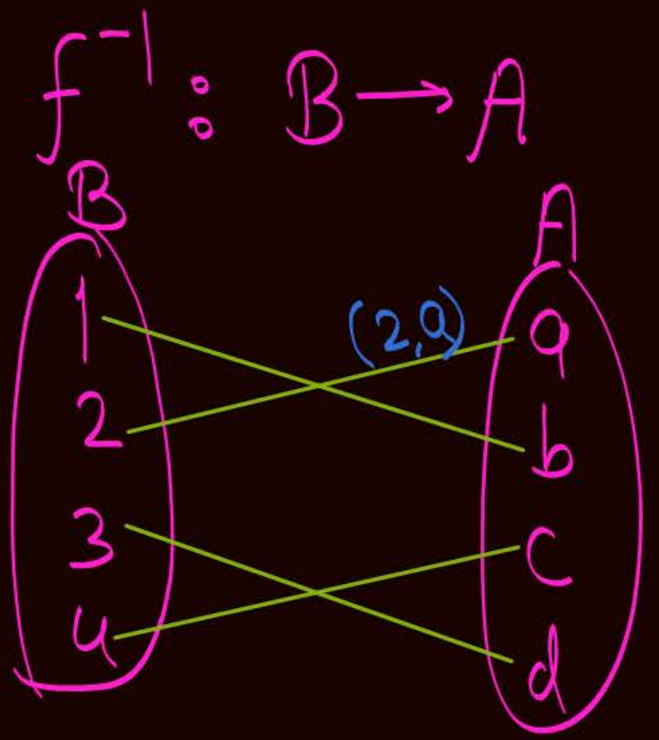
eg:- $A = \{1,2\}$, $B = \{a,b,c\}$
$R = \{(1,a), (2,b), (2,c)\}$

$R^{-1} = \{(a,1), (b,2), (c,2)\}$

Slide

$f: A \to B$





$f(a) = 2$ ⟶ $f^{-1}(2) = a$

$f(b) = 1$ ⟶ $f^{-1}(1) = b$

$f(c) = 4$ ⟶ $f^{-1}(4) = c$

$f(d) = 3$ ⟶ $f^{-1}(3) = d$

Note:-

Let R is a relation on set A, then

Symmetric closure of $R = R \cup R^{-1}$



Topic : Composite of two Relations

Let X, Y and Z be three sets, and

$R : X \to Y$, i.e. $R \subseteq X \times Y$ $\quad$ $S : Y \to Z$, i.e. $S \subseteq Y \times Z$

- If R is a relation from X to Y, and S is a relation from Y to Z, then their composition R;S (or simply 'RS') is the relation defined as,

$$R;S = \{(a,c) \mid (a,c) \in X \times Z \text{ and there exists } b \in Y \text{ such that } (a,b) \in R \text{ and } (b,c) \in S\}$$

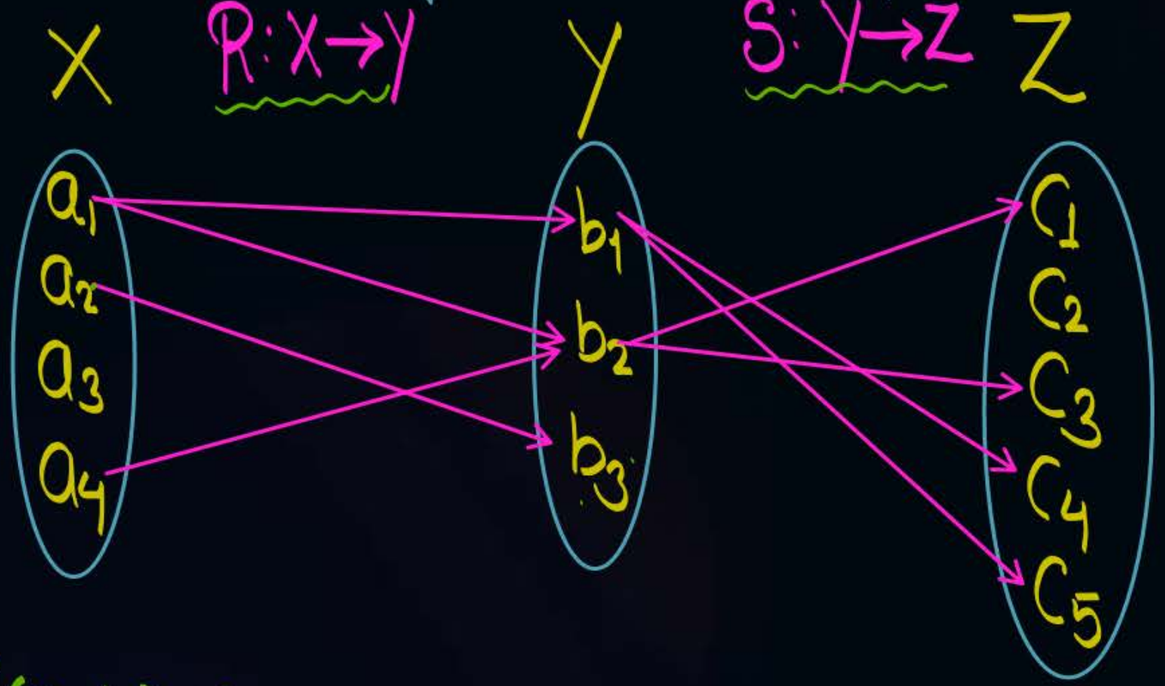
i.e. R;S will be a subset of $X \times Z$, i.e. R;S will be a relation from X to Z

i.e., R;S is a relation from X to Z defined by the rule that $(a,c) \in R;S$ if and only if there is an element $b \in Y$ such that $(a,b) \in R$ and $(b,c) \in S$

Slide

Topic : Composite of two relations

$R \subseteq X \times Y$ $\quad R: X \to Y$ $\quad S \subseteq Y \times Z$ $\quad S: Y \to Z$



$R = \{(a_1, b_1), (a_1, b_2), (a_2, b_3), (a_4, b_2)\}$

$S = \{(b_1, c_4), (b_1, c_5), (b_2, c_1), (b_2, c_3)\}$

Composite of relation R & S i.e. R;S is a relation from X to Z

$R;S : X \to Z$

$R;S \subseteq X \times Z$



$R;S = \{(a_1, c_1), (a_1, c_3), (a_1, c_4), (a_1, c_5), (a_4, c_1), (a_4, c_3)\}$

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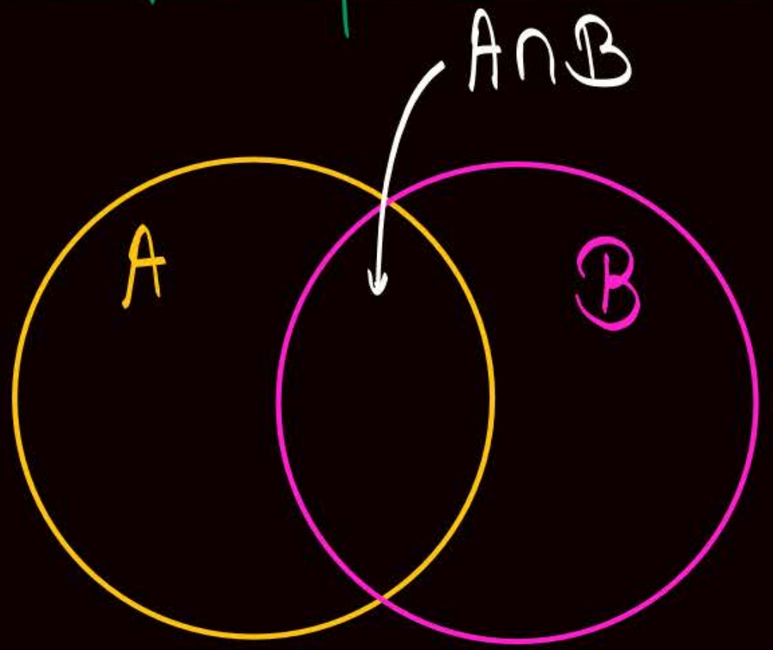
Note: Composite of two relations R and S need not be Commutative

i.e In general. $R;S \neq S;R$

$R: X \to Y$ & $S: Y \to Z$

$R;S: X \to Z$

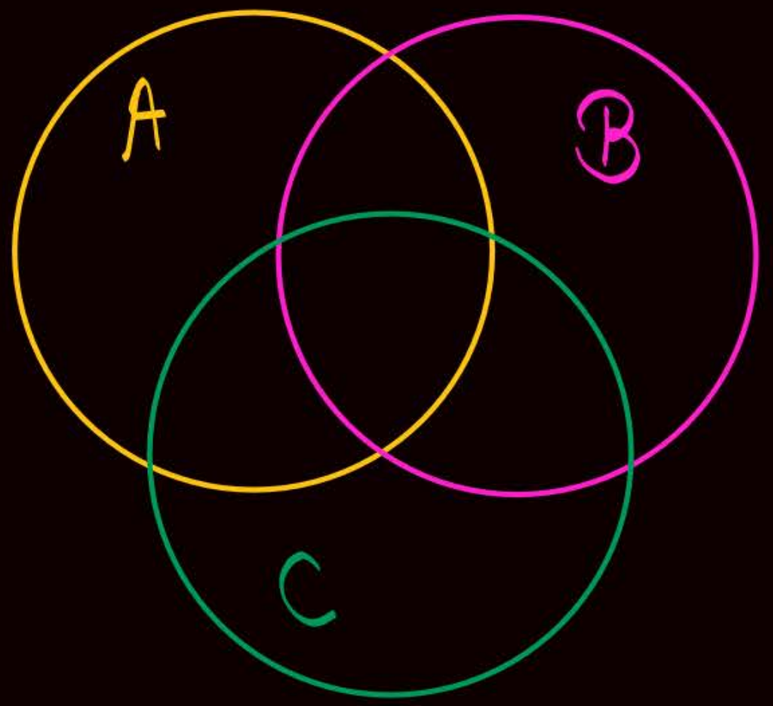
Principle of inclusion & Exclusion :-



$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Principle of inclusion & Exclusion :-



$$n(A \cup B \cup C)$$

$$= n(A) + n(B) + n(C)$$

$$- n(A \cap B) - n(A \cap C) - n(B \cap C)$$

$$+ n(A \cap B \cap C)$$

H.W.

Q. Let X and Y are two sets such that

$|X^c| = 15$

$|Y^c| = 8$

and $|X \cap Y^c| = 5$

then $|X^c \cap Y| = ?$

H.W

Q: The expression $(A \cup B \cup C') \cap ((A' \cap B' \cap C) \cup C')$ is equivalent to:

(a) $(A \cup B) - C'$

(b) $(A \cup B) \cap C$

(c) $(A \cup B) \cap C'$

(d) None of the above

H.W.

MSQ:- Let 'P' be the set of all human beings, and R is the relation defined on set 'P' by the rule $(a, b) \in R$ if and only if blood group of a is same as blood group of b

then Relation R is

(a) Reflexive
(b) Symmetric
(c) Transitive
(d) Anti-symmetric

H.W MSQ: Let 'P' be the set of all human beings, and R is the relation defined on set 'P' by the rule
$(a, b) \in R$ if and only if 'a' is brother of 'b'

then Relation R is

(a) Reflexive
(b) Symmetric
(c) Transitive
(d) Anti-symmetric
(e) None of the above



2 mins Summary

- Topic: Transitive closure
- Topic: Complement of a relation
- Topic: Inverse of a relation
- Topic: Composite of two relations

Slide

THANK - YOU